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START-UP SUCCESS PREDICTION

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Abstract: Startups must regularly evaluate their progress towards their goals. It is important to know the success rate of a firm because the failure rate of startups was about 90% in 2019. Initial public offerings (IPOs) and mergers and acquisitions are two common paths to success for companies. In this study, we explore what factors contribute to a startup's chances of being acquired or merged. Five models have been developed and compared using available historical data on startups to determine if a business would be purchased. Decision Trees, Random Forest, Gradient Boost, Logistic Regression, and MLP Neural networks are some of the models that have been tried. Important characteristics, including as valuations, fundraising rounds, investments, etc., are included in the data used to train these models. One could predict the future of the company with the use of these models. As we used the models, we improved accuracy to roughly 92% across the board. Potential investors and other firm stakeholders might benefit greatly from having access to this data.

Keywords: COVID, UML, IPO, Random Forest

I. INTRODUCTION

Startups have been getting a lot of press lately all across the world. New businesses appear to be popping up every minute. Startups and entrepreneurship have gained popularity in recent years, particularly after the global pandemic. As a direct result of the pandemic, this year has seen a phenomenal increase in the number of new enterprises established around the world. This uptick in new businesses can be traced back to the legions of laid-off workers who decided to go it alone. In July of 2020, a record 551 657 applications were filed to begin new businesses in the United States. The Census Bureau reports that this is a 95% increase from the same time in 2018.

The importance of startups in driving economic expansion and job creation is becoming more widely acknowledged. The startup culture has become counterproductive to societal progress. Because of their capacity to produce useful answers, they serve as catalysts for societal and economic change by way of technological innovation and scalability. People now have access to a new revenue stream because to the rise of startups. During the COVID period, unemployed people accounted for one in five

new micro business starts. It was 12% before the COVID outbreak. This demonstrates how micro businesses and startups are helping a vast number of people break the cycle of poverty. Due to their ability to inject new vitality and competition into the economy, startups are widely regarded as the most dynamic economic organizations operating today. As a result, the economy as a whole remains robust and productive, yet individual businesses have a more difficult time resting on their laurels.

While many new businesses have been formed in recent years, not all of them have been successful. In 2019, over 90% of new businesses failed. According to the data, over 22.5% of new businesses fail in their first year, followed by 30% in year two, 50% in year five, and 70% by year 10. This is not encouraging news for those who dream of starting their own business. The list of possible explanations for such setbacks is vast. Low consumer interest is a contributing factor. Many new businesses are based on a product for which there was never much of a market to begin with. Financial constraints are another possible explanation. Many new businesses fail because they lack the foundation for long-term success. Some additional causes include bad judgment on the part of managers, natural disasters like Covid, or even fierce rivalry.

As a result, it is crucial for modern business owners to evaluate whether or not they are heading in the right direction. While there is no single formula for startup success, there are two main types of companies that tend to do well. An Initial Public Offering (IPO) is a method through which a company can raise capital by offering shares of stock to the public for the first time. The startup's owners and investors can also cash out quickly by selling their shares to a larger company through a merger or acquisition (M&A). This is often referred to as simply "the plan."

II. RELATED WORKS

More and more universities, governments, and private firms are investing in and encouraging people to follow their ideas through startups, and this has led to a global rise in the number of new businesses. More than half of all startups go bankrupt within the first four years. Nevertheless, the success rate of companies receiving venture capital is only 25%. In order to forecast a startup's financial and managerial viability, I developed a

machine learning model. The study aims, through the application of data mining and machine learning techniques, to develop a predictive model that has as its dependent variable a label to classify whether a startup is (already) successful or not, with a focus on how a start-up or an investor could explore all this knowledge for a better decision making in investment strategy and financial gain. Supervised machine learning algorithms were evaluated for their ability to produce a predictive model, including: A Random Forest. Managerial variables include things like the number of employees, competitors, location, age of the company, founders' backgrounds, burn rate, and various news articles about the company that were scraped off the internet, while financial variables include things like investments in each funding round, valuation after each round of funding, current market value, total funds, investments, and acquisitions by the company, and financial background of key people. CrunchBase presents a wealth of opportunities for research, and the acquisition of companies is a particularly intriguing challenge that could benefit from a method centred on venture capital (or other types of investment) traits and various machine learning algorithms. This encompasses preparatory work, the introduction of new variables, challenges encountered, and their resolutions. Lastly, we will discuss the implementation of machine learning techniques to create the proposed prediction model using supervised learning, including the design of the experiment, the data collected, and our ultimate conclusions.

The global pandemic has brought about a true boom in startups, as the number of new companies around the world has significantly surpassed the indicators of last year. Such a surge in entrepreneurship is being attributed to workers who were laid off and started their own businesses. The Financial Times reports, citing official state statistics, that a boom in entrepreneurship in many countries has been recorded against the backdrop of the coronavirus pandemic of 2020.

In recent years, startups have been receiving increased attention in many parts of the world. In India, the number of startups has increased fast and more support has become available in all dimensions. This paper analyses the current state of the Indian startup ecosystem and has three goals: to provide an understanding of the growth drivers and motivations of Indian startup founders; identify challenges facing these startups; and outline the pillars in place that support them. The analysis uses data collated from semi-structured interviews with startup founders, investors, and representatives of support organisations. In addition, a survey of relevant literature strengthens the robustness of the findings.

A significant factor in this new business boom is the growing number of entrepreneurs from diverse backgrounds. Website giant GoDaddy has partnered with a number of universities on its Venture Forward research project to learn more about the people behind this change and the types of companies they are starting. Many of the new firms fall into the category of microbusinesses. Venture Forward estimates that there are 45 million microbusinesses in the US. Around half of these are run by a single person – or solopreneurs – and around 90% have fewer than 10 employees.

Any entrepreneur knows that, in your early days, no one will hesitate to tell you that most new businesses fail. It's annoying, but they're right. However, what's usually missing in the message is the why. Instead, there's a sense of inevitability. Starting a business is hard, so of course, most of them fail, right?

But in my experience, there are specific reasons startups most often fail. Recognizing them and preparing for them dramatically shifts the odds in your favor. I've built my business around

partnering with healthcare entrepreneurs to avoid these pitfalls — so far with a record of 6-0, with three new ideas coming to life. Why does one person actually succeed in starting a business, while a second person gives up? In order to answer this question, a sample of 517 nascent entrepreneurs (people in the process of setting up a business) was followed over a three-year period. After this period, it was established that 195 efforts were successful and that 115 startup efforts were abandoned. Our research focuses on estimating the relative importance of a variety of approaches and variables in explaining pre-startup success. These influences are organized in terms of Gartner's (Academy of Management Review 10(4), 696–706 [1985]) framework of new venture creation. This framework suggests that start-up efforts differ in terms of the characteristics of the individual(s) who start the venture, the organization that they create, the environment surrounding the new venture, and the process by which the new venture is started. Logistic regression analyses are run for the sample as a whole as well as for subgroups within the sample, namely for those with high ambition vs. low ambition and for those with substantial vs. limited experience. The results point to the importance of perceived risk of the market as a predictor of getting started vs. abandoning the startup effort.

III.EXISTING SYSTEM:

The current body of research is mostly concerned with forecasting the success rates of established businesses. Yet, there are major differences between the success predictions of corporations and those of startups, which renders the currently available models ineffective for predicting the success of startups. When it comes to making decisions in such a high-risk environment, the actors involved in the startup ecosystem can tremendously benefit from a quantitative strategy. This is because the processing of enormous volumes of data requires a lot of energy as well as a lot of time.

IV.PROPOSED SYSTEM:

In the system that we have proposed, we have eliminated the unnecessary data by making use of the productive packages in Python. Additionally, we have manipulated the data in such a way that the system will be able to predict the data in a fraction of a second, which will also increase the system's overall efficiency.

A revised definition of the target variable will be included in it. The influence of venture capital is the primary topic of the new feature. The user is given the opportunity to engage in conversation with the web page through the inclusion of a dynamic component in the system.

The research carried out by Pan et al. is wholly predicated on the application of machine learning, and the data set was acquired using CrunchBase. The primary objective is to make a prognosis regarding the status of start-ups, taking into account whether or not these businesses have undergone M&A (mergers and acquisitions) or an initial public offering (IPO) (Initial public offering). Among the machine learning techniques that have been applied, the most common ones are logistic regression, random forests, and k-nearest neighbours. In the study, a number of machine learning (ML) algorithms, such as those described above, are investigated to identify which algorithm works best with the dataset. The F1 scores are used as the primary criterion, and it was discovered that the KNN algorithm had the greatest possible score.

Arroyo and his colleagues have conducted research that focuses not only on the two classic classification categories of mergers

and acquisitions and initial public offerings (IPO), but also on other possible outcomes, such as a second investment round or the closure of the company. Support Vector Machines, Decision Trees, Random Forests, Very Randomized Trees, and Gradient Tree Boosting are some of the machine learning methods that have been utilised in this investigation. Once again, the well-known database Crunchbase was utilised in the course of this inquiry. The primary focuses of their strategy are on early stage businesses, time-aware analyses, and the challenge of predicting multiple classes of data simultaneously. According to the results of the research, the best algorithm, which is called Gradient Tree Boosting, has an accuracy of approximately 82% across the world.

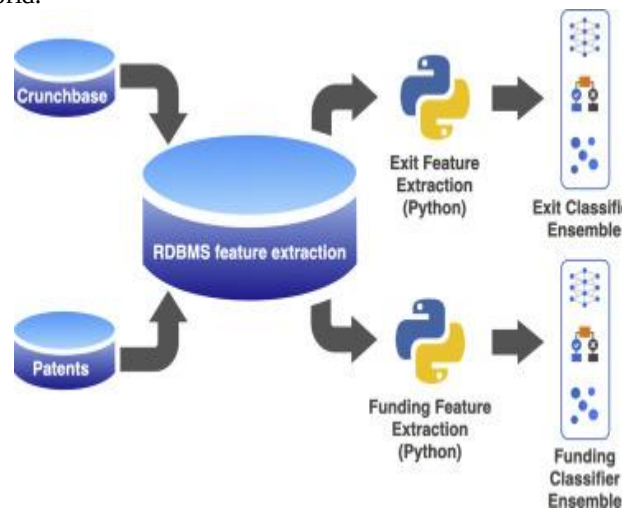


Fig 1: System Architecture

The purpose of a framework for handwriting acknowledgement is to convert an interpretation of input variables into designs that are meaningful to machines. Vehicle tag recognisable proof, postal letter arranging, Check truncation framework (CTS) checking and recorded document conservation in archaeological divisions, old archive robotization in libraries and banks, and other applications are some of the generally standard uses of this technology. These fields manage massive data bases, which necessitates exceptional acknowledgement precision, reduced processing complexity, and predictable acknowledgment framework performance. It is generally accepted that more profound neural models offer more value than more shallow neural models. A convolutional neural organisation is a type of deep neural network that has applications in a variety of domains, including but not limited to the following: face recognition, picture categorization, object recognition, proposal frameworks, signal processing, natural language processing, and computer vision. They are more effective than their predecessors (Multi-layer perceptron (MLP), and so on) due to the fact that they can automatically recognise the fundamental components of a thing (here, an item can be a picture, a written by hand character, a face, and so on) without the need for any human oversight or collaboration. The results of progressive component learning's strong skills can be seen in the form of a random forest with remarkable capabilities.

V. RESULTS AND DISCUSSION

A Random Forest, sometimes known as RF, is a grouping of Decision Trees (DT). In contrast to LR and SVM, RF does not anticipate the presence of linear characteristics, despite operating in a highly multidimensional space. It is possible to think of it as employing bagging on a number of different tree classifiers, in its

most basic form (Leo Breiman, 2001). It is not possible to build multiple trees on the same data because it will get the same results. Therefore, randomness of two different types is introduced: each tree is built on slightly different rows, sampled with repetitions from the original (bagging), and each tree (or in some cases, each branch decision) is built using a randomly selected subset of columns. Nevertheless, it is not possible to build multiple trees on the same data because it will get the same results. The purpose of RF is to prevent overfitting, and it does this by developing smaller (shallower) trees using the random subsets of features that it generates and using these subsets to build the trees.

Although its bias is the same as a single Decision Tree, its variance decreases as we increase the number of trees, which also decreases the chances of overfitting. One of the primary advantages of Random Forests is the way the algorithm handles the Bias-variance trade-off, which is a central problem in supervised learning. Other benefits include the fact that it can operate effectively on large datasets, that it can process thousands of input features without having to delete any of those features, that it can provide estimates regarding which variables are important to the classification, that it can handle missing data, and that it can even keep its high level of accuracy when this proportion is relatively high (Leo Breiman, 2001).

For this particular data set, we have utilised three distinct kinds of algorithms, and we have investigated the percentage of total score achieved by each of these method kinds. The variable known as "status" serves as the decisive factor in this data set "column that might take on two different values, acquired and closed. When a startup is "acquired," it indicates that it was successful and was purchased by another company "closed" indicates that the startup was unsuccessful. Just the columns that contribute significantly to the overall value of this dataset were retained by us. Certain values, such as the date or the state id, were deemed to be irrelevant, and as a result, they were eliminated. We have maintained the data set with the necessary columns intact. The dataset was then divided in half as a result of this step. One is the test data, while the other is the train data. We have used 25% of the data in the set as test data, and we have used the remaining 75% as train data.

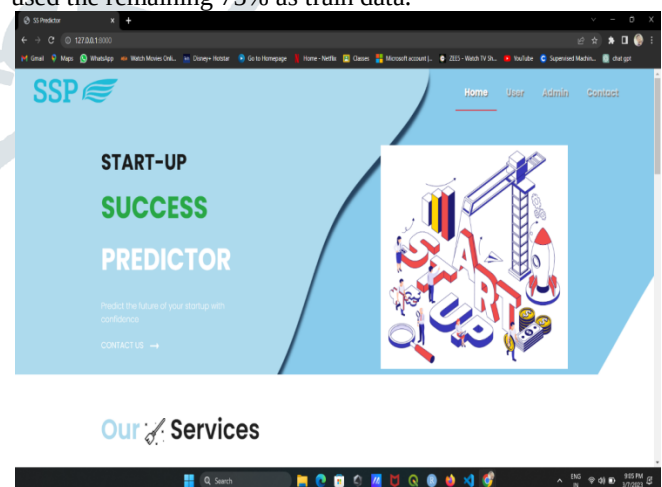


Fig 2: Initial web page

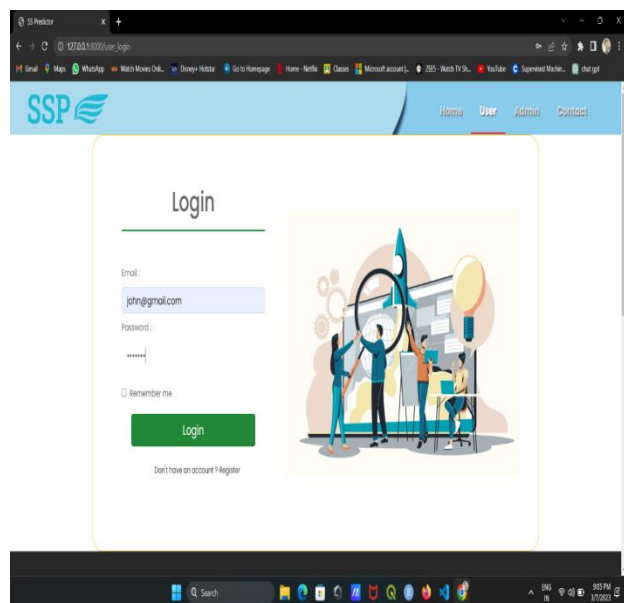


Fig 3: login page

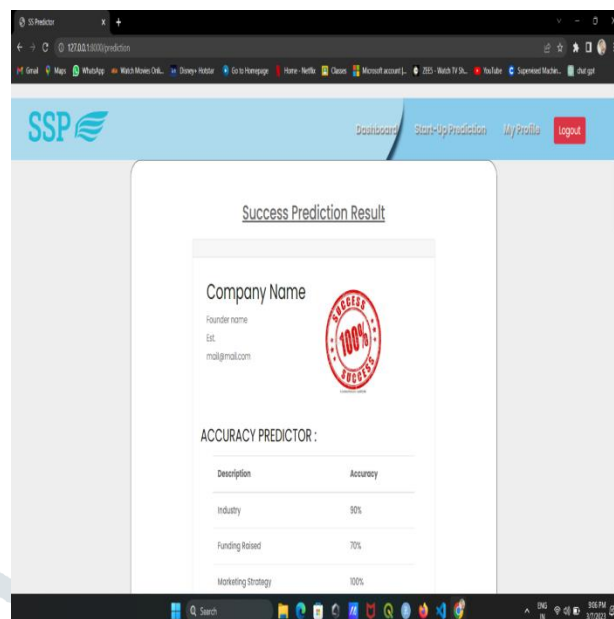


Fig 6: Results

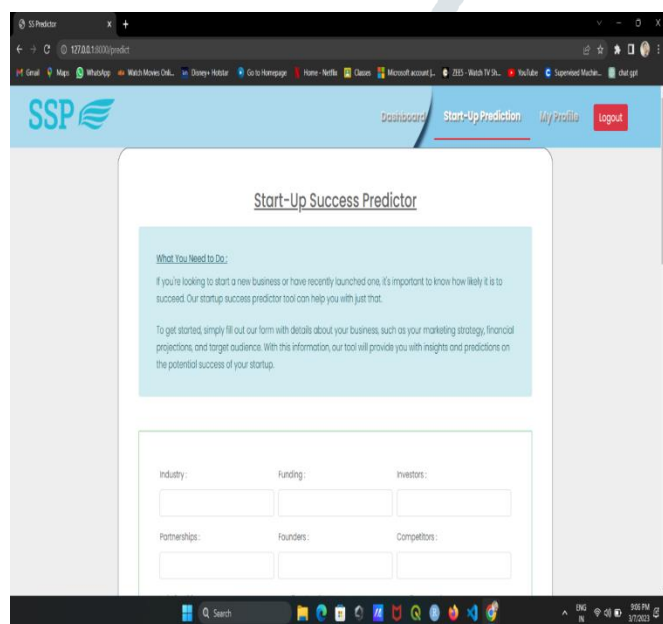


Fig 4: Startup success prediction page

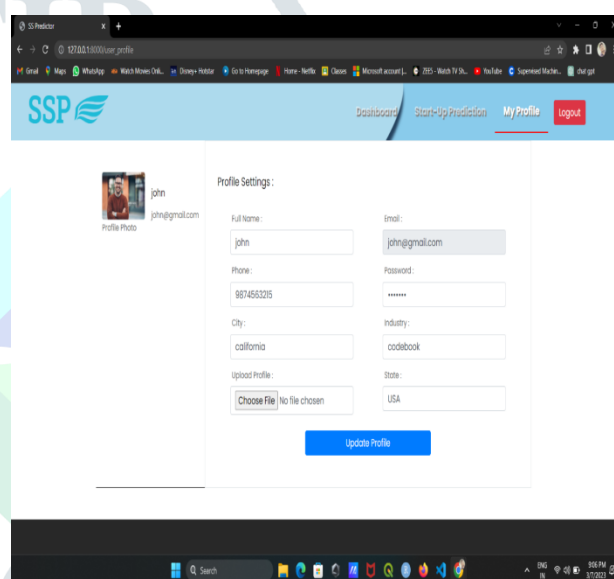


Fig 7: Profile settings

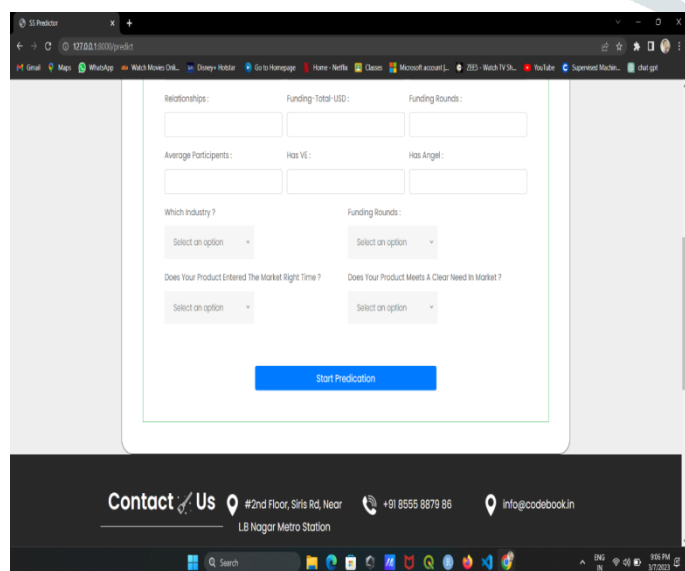


Fig 5: Here we have to give inputs

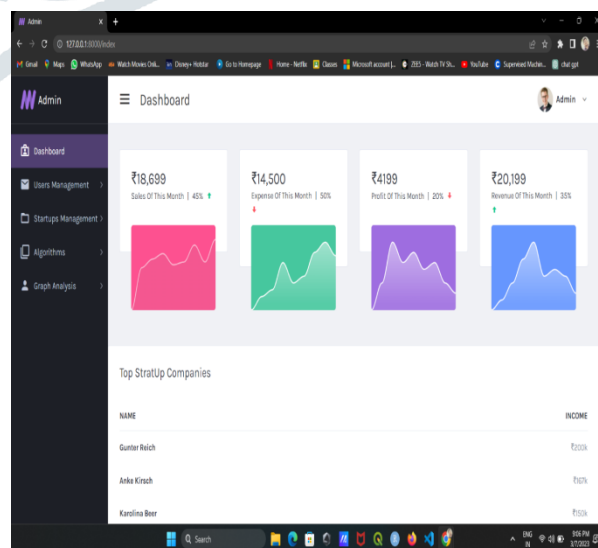


Fig 8: Dashboard to check overall company details

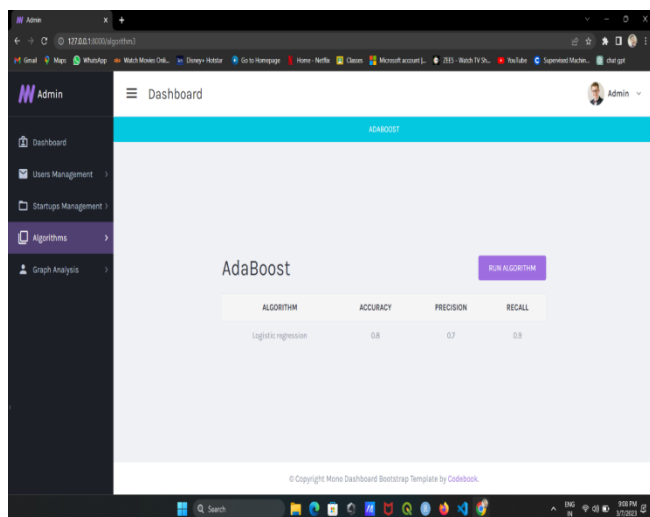


Fig 9: Checking the accuracy of the algorithms

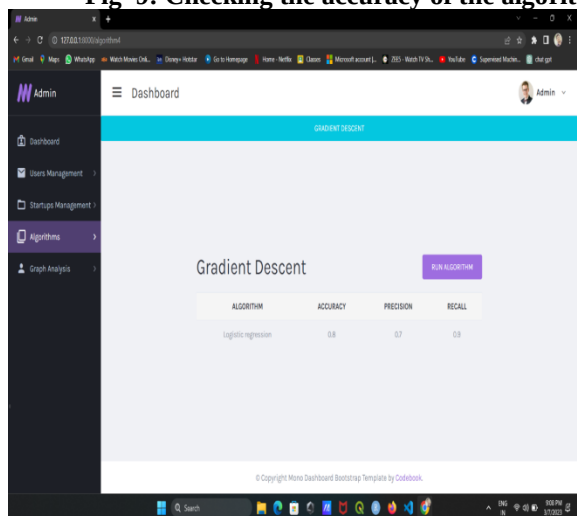


Fig 10: Checking the accuracy of the algorithms

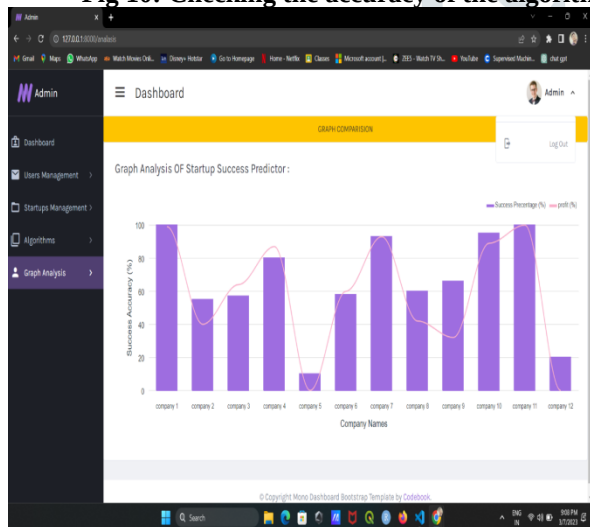


Fig 11: Graph analysis of companies

VI. CONCLUSION

Success rate predictions for new businesses are the focus of this research. The abundance of published works on the topic of successful startups emphasized the need for additional study. Success rates of well-established businesses are the primary focus of the current body of research. Existing models are unhelpful for forecasting the success of startups due to the large gaps between the success rates of established companies and new ventures. Due to the high risk nature of the startup ecosystem and the time and effort required to evaluate large amounts of data, a

quantitative approach can be quite useful for the various actors involved.

We have built models for early-stage startup success/failure prediction using many machine learning methods. The relevant models have attained precision accuracies of 92.4%, 92.3%, 92.6%, 91.9%, and 91.8%. We have also analysed the ROC area values. Given the high quality of our predictions, we are confident in saying that our prediction models can be used by any early stage startup (at each milestone) to forecast their future success. Our research also shows that the aforementioned characteristics (TABLE I) are highly correlated with a company's likelihood of success as a startup. To ensure the company's success, the core committee should have members with both broad and deep expertise in the industry and finance.

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